MIL-C-45420G(AR)
31 JULY 1981
SUPERSEDING
MIL-C-45420F(PA)
1 October 1976

MILITARY SPECIFICATION

CAP, BLASTING, ELECTRIC, M4, ASSEMBLY, OR CAP ASSEMBLY, BLASTING, CONNECTOR AND ELECTRIC

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not covered by the drawings, and provides quality assurance provisions for the parts and assembly for two types of blasting caps designated as:

a. Type I - Cap, Blasting, Electric, M4, Assembly

*b. Type II - Cap Assembly, Blasting, Connector and Electric

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

W-B-411	Blasting Machine; Galvanometer. Blasting; Resistor Assembly.
QQ-L-201	Lead, Sheet.
PPP-B-636	Boxes, Fiber.
PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain for Sealing and Securing.

*see (6.6)

FSC 1375

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-P-116 - Preservation, Method of.

MIL-B-2427 - Boxes, Ammunition Packing, Wood, Nailed.

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For.

MIL-F-50449 - Filler, Sheet Form (For use in Ammunition Containers).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-650 - Explosive, Sampling, Inspection Testing.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

PRODUCT AND PACKAGING DRAWINGS

8796522 - Marking Diagram and Sealing for Wood Packing Boxes
*8835150-2 - Connector and Electric Blasting Cap Assembly
8836013 - Cap, Blasting, Electric, M4, Assembly

INSPECTION EQUIPMENT DRAWINGS

9201429 - Sinusoidal Pulser
9201430 - Operating Instructions
9201434 - Calibration Procedures

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specification procurement functions, should be

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obtained from the procuring activity or as directed by the Contracting Office.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the data of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

TITLE 49 - Transportation, Parts 100-199.

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402. Orders should specify, "49 CFR 100-199 (latest revision)".)

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings, specifications and standards.

3.2 Blasting caps. The electric blasting cap assemblies shall comply with all requirements specified on Drawing 8836013 or 8835150-2, as applicable, with all associated drawings, and all requirements specified in applicable specifications and standards.

3.3 Moisture content of the lead azide. The mositure content of the lead azide at the loading station, at the time of loading the blasting cap assemblies, shall not exceed 0.50 percent when tested as specified in 4.5.1.

*3.4 Waterproofness. The blasting cap assemblies (See Dwg. 8836013 or 8835150-2, as applicable) shall function after submersion for 4.0 plus 0.25 hours at a temperature of 70 plus or minus 5 degrees Fahrenheit and at a pressure of 15.0 plus 0.50 pounds per square inch, when tested as specified in 4.5.2.

3.5 Non-functioning. The blasting cap assemblies shall not function when subjected to the test specified in 4.5.5.

3.6 Functioning. The blasting cap assemblies shall function after conditioning when tested as specified in 4.5.6.

3.7 Workmanship. All parts and assemblies shall be fabricated and finished (or fabricated and loaded) in a thorough, workmanlike manner, and all manufacturing, processing, and assembly operations shall be correctly

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performed. They shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease and oil (except where specifically required), corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any part or assembly nor shall they be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. Required packing shall be dry.

3.8 Wire and pin solder connection. The solder connection between wire and pin prior to molding shall withstand a static load of 1.50 plus or minus 0.10 pounds without separation, when tested as specified in 4.5.7.

3.9 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provision. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item.

a. First Article Inspection.
b. Quality Conformance Inspection.

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2 and 4.3.3. The first article sample shall consist of the following items in sample quantities as indicated in Table I.

4.3.2 Inspections to be performed. See MIL-A-48078 and paragraphs 4.4.2.1 thru 4.4.2.13, testing per paragraph 4.4.3, molding inspections per paragraph 4.4.5, and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE 1 - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-45420G(AR)

PARAGRAPH *	TITLE: Cap, Blasting, Electric, M4 Assembly or Connector and Electric Blasting Cap Assembly		SHEET 1 OF 2		DRAWING NUMBER: See Below NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Spool Dwg. 9227086 (Type 1 only) Examination for Defects	5(a)		3.2	4.4.2.1
	Plug Assembly Dwg. 9260223 with a minimum 6" of wire Examination for defects Pull Test	5(a) 5		3.2 3.8	4.4.2.2 4.5.7
	Female connector & Shorting Plug Dwg. 8800910 with a minimum of 6" of wire Examination for Defects, Pull Test Pull Test	5(b) 5(b)		3.2 3.2	4.4.2.7 as applicable 4.5.3
	Female connector & Shorting Plug Dwg. 8800910 with a minimum 6" of wire Examination for Defects	5(a)		3.2	4.4.2.7 as applicable
	Female Contacts Dwg. 8863763 Examination for Defects	25		3.2	4.4.2.6
	Shorting Prong Dwg. 8861595 Examination for Defects	25		3.2	4.4.2.5
	Sleeves Dwg. 8885031	25		3.2	Inspect to Dwg. 8885031
	Cup 8830953-1 or 8830953-2 Examination for Defects	25		3.2	4.4.2.3

NOTES:

(a) Number indicated is per mold cavity. Items shall be accurately identified from each mold cavity.
(b) Prior to potting.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
*	Cap, Blasting, Electric, M4 Assembly or Connector and Electric Blasting Cap Assembly	2 OF 2	See Below
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cap, Blasting, Electric, M4, Assembly				
	Dwg. 8836013				
	Examination for Defects	50		3.2	4.4.2.9
	Resistance of Assembly	50		3.2	4.5.4
	Non-Functioning	50		3.5	4.5.5
	Waterproofness (a)	25		3.4	4.5.2
	Function at Cold Temperature (-25°F)	25		3.6	4.5.6
	Connector and Electric Blasting Cap Assembly				
	Dwg. 8835150-2				
	Examination for Defects	50		3.2	4.4.2.9
	Resistance of Assembly	50		3.2	4.5.4
	Non-functioning	50		3.5	4.5.5
	Waterproofness (a)	25		3.4	4.5.2
	Function at Cold Temperature	25		3.6	4.5.6
	Wire and connector assembly				
	Dwg. 9275976	25		3.2	4.4.2.8
	Examination for defects				
	Connector and Electric Blasting Cap Assembly				
	Dwg. 8835150-2	25		3.2	4.4.2.4
	Examination for defects				

NOTES:
(a) After water immersion, assemblies shall be functioned statically against the lead disc and 10 shall be functioned in series.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. (See MIL-A-48078). Unless otherwise specified in the classification of defects and test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.4.4.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-45420G(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9227086
*4.4.2.1	Spool				NEXT HIGHER ASSEMBLY 8836013
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Length, max.		0.65%	3.2	Gage
202	Width, max.		0.65%	3.2	Gage
203	Height, max.		0.65%	3.2	Gage
204	Evidence of poor workmanship		1.0%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-45420G(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.2	Plug Assembly				9260223
					NEXT HIGHER ASSEMBLY 8835150-2
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of plug (a)		0.40%	3.2	Gage
102	Length of plug, min. (a)		0.40%	3.2	Gage
103	Bridgewire improperly soldered		0.40%	3.2	Visual
104	Pull Test prior to molding		- - -	3.8	4.4.3.2. 4.5.7
Minor					
201	Length from plug to bridgewire, min.		.65%	3.2	Gage
202	Evidence of poor workmanship		1.0%	3.7	Visual

NOTES:
(a) See 4.4.5 for alternative method of inspection.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-45420G(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.3	Cup				8830953
					NEXT HIGHER ASSEMBLY 8835150-2
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Cup split or cracked		100%	3.2	Visual
Major					
101	Thickness of bottom		0.40%	3.2	Gage
102	Concavity of bottom, max.		0.40%	3.2	Gage
Minor					
201	Inside diameter, min.		0.65%	3.2	Gage
202	Outside diameter, max.		0.65%	3.2	Gage
203	Length		0.65%	3.2	Gage
204	Evidence of poor workmanship		1.0%	3.7	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-45420G(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.4	Connector and Electric Blasting Cap Assembly				835150-2
					NEXT HIGHER ASSEMBLY 8836013 or 9244404
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length of cap, max.		0.40%	3.2	Gage
102	Outside diameter at crimp end, max.		0.40%	3.2	Gage
103	Crimp missing or does not extend 360 degrees		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.7	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-45420G(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
*4.4.2.5	Shorting Prong	1 OF 1	8861595
			NEXT HIGHER ASSEMBLY 8800910

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length		0.40%	3.2	Gage
102	Diameter of ends		0.40%	3.2	Gage
103	Distance between centers of ends		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmenship		1.0%	3.7	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-45420G(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.6	Contact, Female				8863763
					NEXT HIGHER ASSEMBLY
					8800910
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
101	Inside diameter of hole before crimp		0.40%	3.2	Gage
102	Inside diameter of hole after crimp		0.40%	3.2	Gage
103	Depth of hole before crimp		0.40%	3.2	Gage
104	Depth of crimp hole		0.40%	3.2	Gage
Minor					
201	Inside radius missing		0.65%	3.2	Visual
202	Knurl incorrect or missing		0.65%	3.2	Visual
203	Chamfer missing		0.65%	3.2	Visual
204	Evidence of poor workmanship		1.0%	3.7	Visual

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
*4.4.2.7	Female connector with shorting plug	1 OF 1	8800910
			NEXT HIGHER ASSEMBLY 9275967

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Bare wire showing between cable and female contact over max. (prior to molding in connector)		0.40%	3.2	Visual
102	Incorrect trim at wire ends (prior to crimping on contact and molding in connector)		0.40%	3.2	Gage
103	Pull Test		100%	3.2	4.5.3
104	Depth of top end of female contacts		0.40%	3.2	Gage
105	Centerline distance between female contacts		0.40%	3.2	Gage
106	Centerline distance between ends of shorting prong		0.40%	3.2	Gage
107	Distance from female contact centerline to outside diameter of plug		0.40%	3.2	Gage
108	Distance from shorting prong centerline to inside radius of dust cap		0.40%	3.2	Gage
109	Depth to outer end prong		0.40%	3.2	Gage
110	Depth of dust cap		0.40%	3.2	Gage
111	Contour of connector, contact end to shoulder		0.40%	3.2	Gage
112	Length of connector, contact end to shoulder		0.40%	3.2	Gage
113	Presence of rubber or other foreign material in entrance end, slot or cavity or contact surfaces		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.0%	3.7	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-45420G(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.8	Wire and connector assembly				9275967
					NEXT HIGHER ASSEMBLY 9260223
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Shorting prong missing, damaged or loose		0.40%	3.2	Visual-Manual
102	Connector damaged		0.40%	3.2	Visual
103	Wire insulation damaged		0.40%	3.2	Visual
Minor					
201	Color of wire		0.65%	3.2	Visual
202	Length of wire		0.65%	3.2	Gage
203	Evidence of poor workmanship		1.0%	3.7	Visual

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PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
*4.4.2.9	Cap, Blasting, Electric, M4, Assembly or Connector and Electric Blasting Assembly	1	1	8836013 or 8835150-2 as applicable
				NEXT HIGHER ASSEMBLY 8835166 or 9244404

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	Non-functioning test		100%	3.5	4.5.5
Major					
101	Any component damaged or missing		0.40%		Visual
102	Waterproofness Test	55/0/1		3.4	4.5.2
103	Resistance Test		100%	3.2	4.5.4
104	Functioning Test	40/0/1		3.6	4.5.6
Minor					
201	Marking misleading or unidentifiable		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.7	Visual

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.10	Carton, sealed				See (5.1)
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Contents loose		0.40%	5.1.1	Manual
102	Carton damaged so that contents are exposed or liable to be exposed		0.40%	5.1.1	Visual
103	Closing for shipment improper		0.40%	5.1.1	Visual
Minor					
201	Marking misleading or improper		0.65%	5.3.1	Visual

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.11	Fiberboard Box, prior to sealing				See 5.1
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Components missing		0.40%	5.1.1	Visual
102	Packing materials (pads or cell partitions) missing, to the extent that contents are loose		0.40%	5.1.1	Visual-Manual
Minor	None defined				

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.12	Sealed packed barrier bag prior to packing in box				See 5.1
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Waterproofness of bag destroyed by rupture, puncture, separation or incomplete sealing		0.40%	5.1.1	Visual
Minor					
201	Bag seal incorrectly located		0.65%	5.1.1	Visual

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.13	Box, sealed wooden packing				See 5.2 and 5.3
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Strapping missing, broken or loose		0.40%	5.2.1	Visual-Manual
102	Board broken or with split terminating at edge		0.40%	5.2.1	Visual
103	Contents of box exposed		0.40%	5.2.1	Visual
104	Contents loose when box is tilted		0.40%	5.2.1	Manual
Minor					
201	Car seal missing, unsealed or improperly positioned		0.65%	5.2.1	Visual
202	Marking misleading or unidentifiable		0.65%	5.3.1	Visual

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4.4.3. Testing. Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.3.1 Moisture content of lead azide. The contractor shall provide adequate controls to assure compliance with the requirements and shall test for verification, a minimum of one sample of each material selected at the loading station at the time of loading from each eight hours production. A composite sample shall not be used. The tests shall be performed as specified in 4.5.1. Failure of the sample to comply with the applicable requirements shall cause the explosive mixture represented by the sample, including any blasting caps loaded with the non-conforming explosive mixture, to be rejected. The contractor shall maintain adequate records to provide identification of the specific amount of explosive mixture and the loaded items which are represented by the sample.

4.4.3.2 Wire and pin solder connection. Prior to molding, 10 soldered wire and pin connections shall be selected from each two (2) hours of production for this test. If any connection fails to comply with the requirement, the entire quantity represented by the sample shall be subjected to 100 percent testing for this defect. All defective connections shall be removed from the two (2) hour production quantity. The test shall be performed as specified in 4.5.7 (Non-destructive test).

4.4.4 Inspection equipment. The inspection equipment required to perform the inspections and tests prescribed in this specification identified in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1, and the test method paragraphs (See 4.5). The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078, and Section 6.3 herein.

4.4.5 Dimensional control of molded parts. In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, and after a curing time for the parts has been approved (see Note), a sample of three (3) parts (as molded) from each cavity may be dimensionally inspected to qualify a new or reworked cavity for use in production or as a control of each cavity during production.

a. The inspection of the parts shall be performed only when the curing time for the parts has been reached. The inspection for control of each cavity during production shall be performed on the above quantities from each cavity after production of each 5000 parts or at the end of each shift, providing the number of parts produced during that shift is less than 5000 parts. Of the three (3) samples, one (1) sample shall be the last part produced. Parts that fail to comply with dimensional requirements after dimensional stability has been attained shall be classed defective.

b. If any defective parts are found during qualification of the cavity, the cavity producing the defective part will not be used in production. If any defective parts are found when inspection is performed for control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for corrective action.

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c. All cavities formerly removed from production because of some fault may, after reworking, be returned to production providing they pass the qualification test as above. The contractor may request a change of inspection frequency providing he presents objective evidence to substantiate the request to the Contracting Officer

NOTE: In establishing a curing time dimensionally check ten (10) parts from each cavity at periodic intervals (e.g., every 30 min, 60 min, etc.), until dimensional stability is attained. The curing time will be from the time the part comes out of the mold until dimensional stability is attained. The inspection data used in determining the curing time shall be sent to ARRADCOM, Dover, NJ 07801, ATTN: DRDAR-QAT. If there is a change in material, cycle time, or if a cavity is reworked, or a new cavity is used, a new curing time shall be established and approved.

4.5 Test methods procedures

4.5.1 Moisture content of lead azide. The moisture content of the lead azide shall be determined in accordance with Method 101.5 of MIL-STD-650 except that a two (2) gram sample shall be tested and the dish shall be covered with a sheet of polytetrafluoroethylene (See 6.5).

4.5.2 Waterproofness. The blasting cap assemblies shall have the cap with a minimum of twelve (12") inches of wire completely immersed in water (spool and remaining wire need not be immersed - Type I only) at the specified temperature for the specified time and pressure. Immediately upon removal from the water, 15 of the sample caps shall be subjected to static functioning against lead discs. The lead discs, made of 8 pound sheet lead complying with QQ-L-201, Grade B, 0.1345 inch in thickness and 1.22-0.01 inch in diameter shall be centered over a 15/16 inch diameter hole in a metal support. The cap shall be suitably fixed in an upright position, with the base in contact with the unsupported section of the lead disc and fired by the energy supplied by a pulse generator. Test shall be conducted using the equipment and procedures specified on Dwgs. 9201429, 9201430 and 9201434. Observation shall be made for complete detonation as evidenced by perforation of the lead disc. The remaining 40 caps shall be either connected in series and fired with an electric current of 1.5 amperes maximum applied to the circuit, for a period not exceeding 50 milliseconds; or else they shall be connected in series, in groups of 10 each, and fired with a 10 cap capacity blasting machine complying with W-B-411, Type I, Class A, through a total resistance of 32 ohms. Observation shall be made for evidence of complete detonation of the caps.

4.5.3 Wire and female connector assembly. Before molding into connector assembly, the wire shall be held securely and the required force applied to each contact in a smooth gradual manner. Any evidence of separation or slippage of the cable and female connector shall be considered as not conforming to the drawing requirement and shall be classed defective and removed from the lot. (Non-destructive test)

4.5.4 Resistance of assembly. The resistance shall be measured across the contacts of the female connector with the test current not to exceed 20 milliamperes using approved measuring equipment. Any assembly failing to comply with all of the specified requirements shall be classed defective (Non-destructive test).

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4.5.5 Non-functioning. The blasting cap assemblies shall be tested by utilizing pulse producing equipment to apply a pulse having energy of 3.0 milliwatt seconds, plus or minus 1.0 milliwatt seconds for not less than 10.0 or more than 24.0 milliseconds. Any assembly which fails to comply with the applicable requirement shall be classed defective and removed from the lot. Test shall be conducted using the equipment and procedures specified on Dwgs. 9201429, 9201430 and 9201434.(Non-destructive test)

4.5.6 Functioning. The blasting cap assemblies shall be conditioned at minus 40 degrees, plus or minus 5 degrees, Fahrenheit for 24 hours, plus or minus 1 hour. They shall be functioned within 3 minutes after removal from the temperature chamber utilizing pulse producing equipment to apply a pulse of 18.0 plus 1.0 milliwatt seconds for not less than 10.0 or more than 24.0 milliseconds. Observation shall be made for evidence of detonation. Test shall be conducted using the equipment and procedures specified on Dwgs. 9201429, 9201430 and 9201434. (Destructive test)

4.5.7 Wire and pin solder connection. Before molding, the wire shall be held securely and the required force applied to each pin in a smooth, gradual manner. Any assembly which fails to comply with the applicable requirement shall be classed defective and removed from the lot.

5. PACKAGING

5.1 Preservation and packaging

5.1.1 Level A (Type I only). Thirty (30) blasting caps shall be packaged in a fiberboard box complete with plain pads and half-slotted partitions assembled to form 30 individual cells in a 6 by 5 arrangement. A plain pad shall be placed on the bottom of the box and an additional one or more pads shall be placed on top of the partitioned cells as required for a tight pack. The fiber box shall conform to PPP-B-636, Type CF, Class Domestic, Variety SW, Grade 200. Box size shall be approximately 21-3/4 by 12-5/8 by 6-3/8 inches in depth, inside dimensions. The partitioned cells and plain pads shall be constructed of the same test fiberboard as the box. Cell size shall be 4-1/4 by 2 by 6-1/8 inches in depth, inside dimensions and the assembled cells shall slide fit into the fiber box. The pads shall be equal to the inside length and width of the box less 1/8 inch. Boxes shall be closed by application of 2 inch wide gummed, kraft tape, PPP-T-45, Type III, Grade B, over the entire length of the top and bottom center seam of the box with a minimum 2 inch extension of tape onto the end panels. All corners of the fiber box shall be

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blunted and the box inserted into a barrier bag conforming to MIL-B-117, Type I, Class E, Style 1 or 2. All excess air shall be exhausted from the bag and the bag opening closed by heat-sealing. The bag seal shall meet the heat-seal test requirements of MIL-P-116. Bag size shall be approximately 20-1/2 by 31 inches inside dimensions.

*5.1.2 Commercial. The blasting caps shall be preserved and packaged in accordance with Code of Federal Regulations, Title 49, parts 100-199.

5.2 Packing

5.2.1 Level A (Type I only). Each unit package shall be packed in a wooden box conforming to MIL-B-2427 for a Type II, Class 2, Grade A box. One or more fillers, sheet form, conforming to the requirements of MIL-F-50449 shall be placed between the barrier bag and the interior sides and ends of the box and on top of the box contents as required to provide a tight pack. The box size shall be approximately 22-1/2 by 13-1/8 by 7-3/8 inches in depth inside dimensions. Side fillers shall be approximately 22 by 7 inches; end fillers shall be approximately 12-1/2 inches. Boxes shall be closed and strapped in accordance with instructions cited in the appendix of the box specification.

*5.2.2 Commercial. Blasting caps shall be packed in accordance with Code of Federal Regulations, Title 49, parts 100-199.

5.3 Marking

5.3.1 Marking (Level A). Each wooden box shall be marked in accordance with Dwg. 8796522. The DOT marking shall be "DETONATING FUZES, CLASS C EXPLOSIVES - HANDLE CAREFULLY."

5.3.2 Marking (Level C). Each box shall be marked in accordance with manufacturer's normal practice and in compliance with Code of Federal Regulations. The DOT markings shall be: "DETONATING FUZES, CLASS C EXPLOSIVES - HANDLE CAREFULLY.

6. NOTES

6.1 Intended use

a. Type I. The electrical blasting cap, M4, is used as the initiating device for Mine, APERS, M18A1.

*b. Type II. The connector and Electric Blasting Cap Assembly is used as the initiating device for Demolition Kit, Cratering, M180 Assembly.

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6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Provisions for submission of first article samples.
c. Provisions of MIL-A-48078.

*6.3 Submission of inspection equipment designs for approval (See MIL-A-48078). Submit equipment designs as required to Commander, ARRADCOM, ATTN: DRDAR-QAT, Dover, New Jersey 07801.

*6.4 Distribution of ammunition data cards. Distribution of data cards shall include the following: Commander, ARRADCOM, ATTN: DRDAR-QAT, Dover, New Jersey 07801.

6.5 Teflon. A product of E.I. DuPont de Nemours and Company, Wilmington, Delaware, has been found satisfactory for making the dish covers used in the moisture content determination of the lead azide (See 4.5.1).

6.6 Asterisks. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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